Healthcare EHR Analytics: An Interactive Power BI Dashboard for Patient Care, Hospital Operations, and Cost Analysis Power BI Dashboarding project proposal

1. Executive Summary:

This capstone project focuses on building an interactive healthcare dashboard in Power BI using electronic health records (EHR). The main goal is to help hospital administrators and clinical heads track patient care quality, hospital bed utilization, and treatment costs in one place. By connecting different clinical and operational tables into a single model, the project will turn complex medical records into simple, actionable visual metrics.

2. Problem Statement:

Background: Hospitals generate huge amounts of data every day across admissions, treatments, medications, and billing. However, because this data is usually stored in separate files or systems, management struggles to see the full picture—such as how chronic conditions affect hospital stay duration or how insurance coverage impacts patient expenses.

Objective: Build a multi-page Power BI dashboard that monitors key hospital performance indicators, tracks patient treatment paths, and evaluates operational efficiency across departments.

Scope: The project focuses on inpatient and outpatient admissions, chronic disease patterns (like diabetes and hypertension), length of stay (LOS), hospital facility usage, and insurance claim reimbursement rates.

3. Data Sources:

Primary Data: Synthea multi-table Electronic Health Record (EHR) dataset containing interconnected tables: patients.csv, encounters.csv, conditions.csv, medications.csv, procedures.csv, organizations.csv, providers.csv, and payers.csv.

Secondary Data: Standard medical classification benchmarks, including SNOMED-CT diagnosis codes, RxNorm medication codes, and CMS hospital length-of-stay guidelines.

4. Methodology:

Data Integration: Load the raw CSV tables into Power BI via Power Query. Clean timestamps, handle missing values, standardize diagnosis codes, and build a star/snowflake relational schema linking fact tables (encounters, conditions, medications) with dimension tables (patients, providers, organizations, payers).

Dashboard Design: Create structured, easy-to-navigate dashboard pages focused on four areas: Executive Overview, Clinical & Chronic Disease Trends, Hospital Operational Flow, and Financial & Payer Performance.

Interactivity: Add dynamic slicers (by department, date, condition, and facility), drill-through pages for individual patient profiles, tooltips for quick summaries, and DAX measures for real-time calculations.

5. Expected Outcomes:

- An interactive multi-page dashboard that tracks patient admissions, hospital bed stays, and disease trends.
- Clear visibility into which chronic conditions lead to frequent visits and longer hospital stays.
- Better understanding of hospital revenue, insurance coverage ratios, and out-of-pocket costs paid by patients.
- A well-structured data model that can easily handle new monthly hospital records.

6. Tools and Technologies:

- Power BI Desktop: For data modeling, DAX measures, and dashboard visualization.
- Power Query (M): For data extraction, cleaning, and table merging.
- DAX (Data Analysis Expressions): For calculating metrics like Average Length of Stay (ALOS), readmission rates, and reimbursement percentages.
- Excel / CSV: For raw data inspection and initial validation.

7. Risks and Challenges:

- Data Modeling Complexity: Handling many-to-many relationships between patients, multiple diagnoses, and multiple prescribed medications without slowing down report performance.
- Data Cleaning: Dealing with varying date formats, blank fields in discharge notes, and large observation files.
- Visual Clarity: Avoiding clutter on dashboard screens while presenting detailed medical and financial information clearly to non-technical users.

8. Conclusion:

This project will provide a practical, data-driven solution for healthcare decision-makers by turning raw EHR records into clear visual dashboards. By bringing clinical outcomes, hospital operations, and financial metrics together in Power BI, the dashboards will help hospital teams spot bottlenecks, improve resource planning, and make better operational decisions.